

Technical Specifications and Compliance of Goods and related services

Item No.	Name of Goods	Detailed Technical Specification and Standards required
1	Software Development (An AI-driven soil analysis tool for processing input data related to soil nutrition, environmental conditions, and historical trends. The system should farmers with actionable insights to optimize soil fertility and improve crop yield).	Purpose of the Document: The purpose of this Software Requirements Specification (SRS) document is to outline the requirements for the development of I Soil, a soil analyzing AI tool designed for checking soil nutrition, including Nitrogen, Phosphorous, Potassium, Electrical Conductivity (EC), pH, Temperature, and Moisture. The tool will facilitate visualization and analysis of soil data for farmers, enabling them to predict their soil condition and optimize the use of compost fertilizer. Scope of the Project: The I Soil project aims to provide an intelligent platform for farmers to analyze and visualize soil data, predicting nutrient levels and recommending optimal compost fertilizer usage. The system will leverage machine learning techniques to enhance the accuracy of predictions and improve over time. System Overview: * System Description: I Soil is an AI-driven soil analysis tool that processes input data related to soil nutrition, environmental conditions, and historical trends. The system will provide farmers with actionable insights to optimize soil fertility and improve crop yield. * System Features: Soil Nutrition Analysis: Detection and analysis of Nitrogen, Phosphorous, Potassium, EC, pH, Temperature, and Moisture levels in the soil. * Data Visualization: Presenting analyzed data through intuitive visualizations for easy interpretation. * Prediction Models: Machine learning algorithms to predict future soil conditions based on historical data. Fertilizer Recommendations: Recommending compost fertilizer usage based on soil analysis and historical trends. * User Interface: An intuitive and user-friendly interface accessible via web or mobile applications. (cont...)
1	Do	Functional Requirements: * Soil Analysis Module: Accept input data for Nitrogen, Phosphorous, Potassium, EC, pH, Temperature, and Moisture levels. Implement algorithms for real-time soil nutrient analysis. IOT Data will be submitted to the authorized cloud where machine learning facilities are available. Provide a detailed report of soil conditions. * Data Visualization Module: Generate graphical representations of soil nutrient levels. Support historical data visualization for trend analysis. * Prediction Module: Utilize machine learning models to predict future soil conditions. Continuously improve prediction accuracy through model updates. * Fertilizer Recommendation Module:

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		Analyze soil data to suggest optimal compost fertilizer types and quantities. Provide a timeline for applying fertilizer for optimum results. * User Interface: Intuitive dashboard for users to input data and view analysis results. Responsive design for both web and mobile platforms. (cont...)
1	Do	Non-functional Requirements: * Performance: Response time for soil analysis should be within seconds. Scalability to handle a growing number of users and data points. * Security: Secure data transmission and storage. User authentication and authorization mechanisms. * Reliability: The system should be available 24/7 with minimal downtime. Regular backups of data to prevent data loss. * Usability: Intuitive and user-friendly interface. Support for multiple languages for a broader user base. (cont...)
1	Do	Technical Requirements: Technology Stack: Backend: Python, Nodes, React, Flutter or other suitable frameworks Database: Postgre SQL or MySQL and Mongo DB for IOT data Machine Learning: Tensor Flow, Scikit-learn or others Frontend: HTML5, CSS3, Java Script. System Integration: Integration with weather APIs for real-time environmental data. Compatibility with common web browsers and mobile platforms connected with IOT Cloud. Constraints: The system's accuracy depends on the quality of input data. Availability of reliable weather data for accurate predictions. Conclusion: This Software Requirements Specification provides a comprehensive overview of the I Soil project, outlining the key features, functionalities, and technical requirements. It serves as a foundation for the development team to design and implement a robust and user-friendly soil analysis tool for farmers.